System Identification and PID Control of a Time-Delay Process

Abstract

This study investigates reduced-order modeling and feedback control of a fourth-order process with transport delay. A three-parameter first-order model is identified from the step-response tangent, while a four-parameter second-order model is obtained by numerical fitting. Relay feedback supplies an estimate of the ultimate frequency and gain for PID design. Time-domain, frequency-domain, generalized Ziegler-Nichols, and a tuning procedure labeled lambda tuning in the source report are evaluated in Simulink. Manual gain adjustment improves the displayed responses. Smith predictor configurations using full and reduced process dynamics are then examined. The reported evidence supports a qualitative comparison of tracking behavior; numerical performance indices and robustness tests are not provided.

1. Process and modeling approach

The process is represented by the continuous-time transfer function

$$G(s) = 1000\, e^{-2.5s} / (s^4 + 126s^3 + 2725s^2 + 12600s + 1000).$$

The plant has unit DC gain and a transport delay of 2.5 s. The MATLAB scripts sample its unit-step response over 0-60 s at 0.01 s intervals. For the first-order plus dead-time (FOPDT) approximation, the maximum-slope point is located numerically, and a line y = mt + c is fitted locally. The parameters follow from K = y(60), L = -c/m, and T = (K - c)/m - L.

$$G_3(s) = 0.990211\, e^{-2.698647s} / (12.982952s + 1).$$

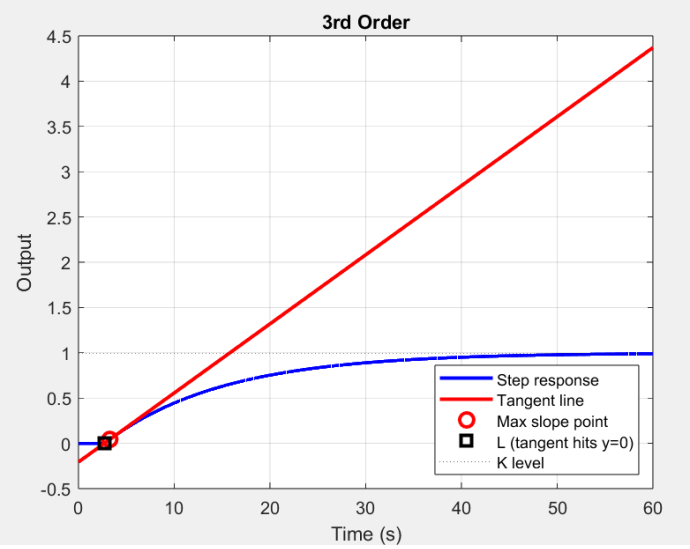


Figure 1. Original step-response and tangent construction for the three-parameter approximation (source report, p. 5). The original plot label refers to the parameter count, not the dynamic order.

The estimated delay includes the physical delay and an approximation of the faster plant dynamics. Because the script uses the response at 60 s as its gain estimate, K = 0.990211 is a finite-horizon approximation rather than the exact steady-state gain of one.

2. Four-parameter identification

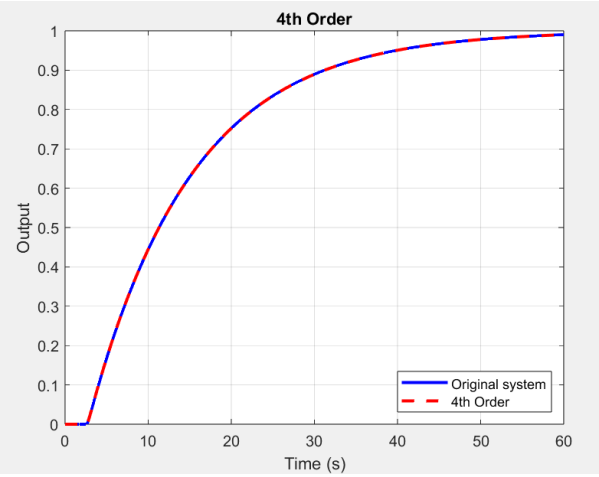
A second-order plus dead-time (SOPDT) model retains two distinct time constants:

$$G_4(s) = K\,e^{-Ls} / [(T_1 s + 1)(T_2 s + 1)].$$

The MATLAB implementation minimizes the sum of squared differences between the original and modeled step responses using *fminsearch*. Optimization is performed in logarithmic coordinates to maintain positive parameter values. The initial estimates are derived from the FOPDT tangent construction, with its time constant divided between two initial lags.

Model	Gain K	Time constant(s), s	Delay L, s
FOPDT	0.990211	T = 12.982952	2.698647
SOPDT	0.999998	T_1 = 12.380540; T_2 = 0.167542	2.551851

The larger SOPDT time constant captures the dominant slow response, while the smaller time constant represents a faster initial component. The fitted delay is closer to the physical value than the FOPDT estimate. The source figure shows close agreement between the SOPDT and original responses over the sampled interval, although no numerical fit-error statistic is reported.



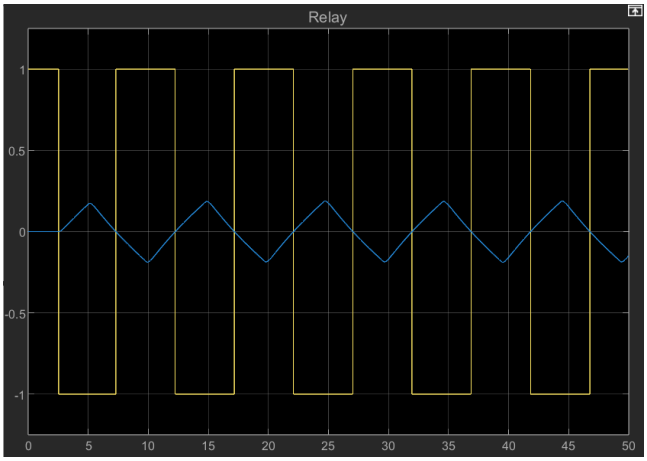


Figure 2. Left: original and fitted SOPDT step responses (p. 8). Right: relay-feedback oscillation, showing the relay signal and process output (p. 11).

3. Ultimate-point estimation by relay feedback

A symmetric relay with output levels +1 and -1 is placed in a negative-feedback loop around the delayed plant at zero reference. The measured process-output extrema are approximately +0.189 and -0.192, giving an oscillation half-amplitude a = 0.1905. The relay half-amplitude is d = 1, and the measured period is T_u = 9.806 s.

$$|G(j\omega_u)| \approx \pi a/(4d) = 0.1496; \quad \omega_u = 2\pi/T_u \approx 0.6407 \text{ rad/s}.$$
$$K_u \approx 1/|G(j\omega_u)| = 6.6845.$$

These are relay-based estimates under the describing-function approximation. At the ultimate point, the process frequency response lies approximately on the negative real axis; 0.1496 denotes its magnitude. The resulting K_u and T_u are used for the frequency-domain tuning calculations.

4. Classical Ziegler-Nichols PID design

The controller is written in ideal form as $C(s) = K_p[1 + 1/(T_i s) + T_d s]$. The corresponding parallel gains are $P = K_p$, $I = K_p/T_i$, and $D = K_p T_d$. The supplied Simulink model uses a derivative filter coefficient $N = 10$. Calculated starting gains and the final manually adjusted gains are distinguished below.

4.1 Time-domain tuning

Using the reaction-curve quantities $a = 0.2058$ and $L = 2.698647$ s, the report applies $K_p = 1.2/a$, $T_i = 2L$, and $T_d = L/2$. These give $K_p = 5.8309$, $T_i = 5.3973$ s, and $T_d = 1.3493$ s. The initial response is oscillatory, with pronounced overshoot. Manual gain adjustment yields a visibly smoother response.

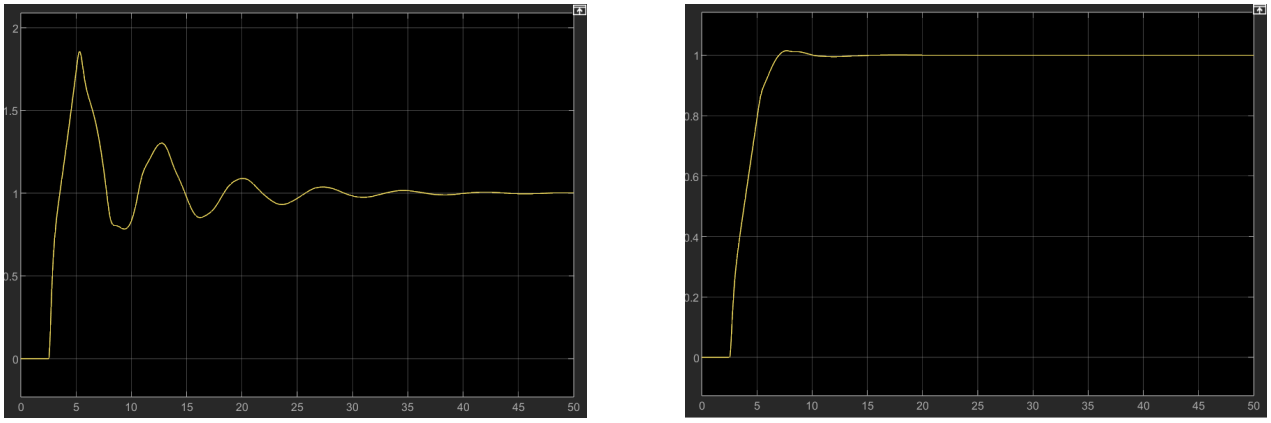


Figure 3. Time-domain tuning: calculated gains (left, p. 14) and manually adjusted gains (right, p. 15).

4.2 Frequency-domain tuning

The ultimate-point rules $K_p = 0.6K_u$, $T_i = T_u/2$, and $T_d = T_u/8$ give $K_p = 4.0107$, $T_i = 4.9029$ s, and $T_d = 1.2257$ s. The initial simulation again exhibits overshoot and damped oscillation. Adjustment reduces these transients, although the plots end below unity; the source does not resolve this apparent tracking discrepancy.

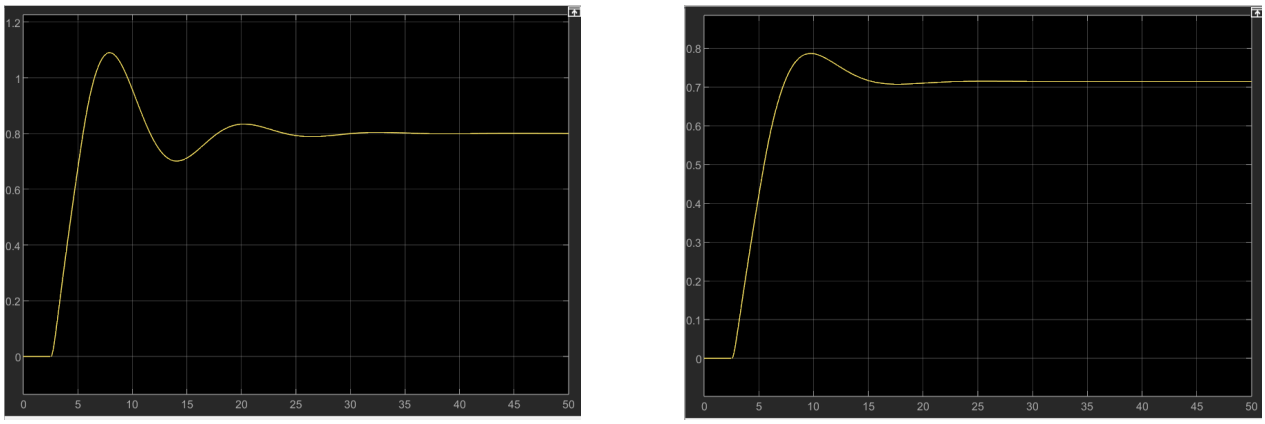


Figure 4. Frequency-domain tuning: calculated gains (left, p. 17) and manually adjusted gains (right, p. 18).

Tuning procedure	Calculated (P, I, D)	Adjusted (P, I, D)
Time-domain Ziegler-Nichols	(5.8309, 1.0803, 7.8675)	(3.3, 0.25, 2.7)
Frequency-domain Ziegler-Nichols	(4.0107, 0.8180, 4.9159)	(2.5, 0.4, 3.5)

The adjusted gains agree with the stored PID parameters. The figures indicate reduced oscillation, but do not establish unit-reference tracking in every case. Exact settling times and overshoot percentages are not tabulated.

5. Generalized tuning and reported lambda tuning

5.1 Generalized Ziegler-Nichols tuning

The generalized design specifies $r_b = 0.41$ and a phase angle of 61°, with the derivative-to-integral time ratio $\alpha = 0.25$. Using the relay estimates, the reported relationships are

$$K_p = K_u r_b \cos(\phi_b), \quad T_i = (T_u/\pi)[(1 + \sin \phi_b)/\cos \phi_b], \quad T_d = \alpha T_i.$$

The resulting values are $K_p = 1.3286$, $T_i = 12.0692$ s, and $T_d = 3.0173$ s. The initial response approaches the reference relatively slowly. Manual adjustment produces a faster displayed approach with a small visible overshoot.

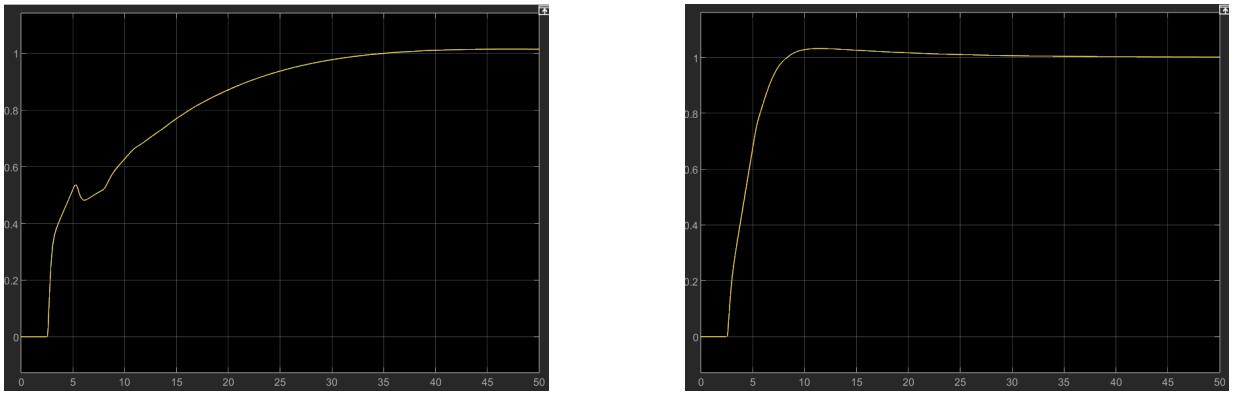


Figure 5. Generalized tuning: calculated gains (left, p. 20) and manually adjusted gains (right, p. 21).

5.2 Procedure labeled lambda tuning in the report

For the identified FOPDT model, the report uses the following relationships, with $q = L/T$:

$$K_p = [T/(KL)](4/3 + q/4),$$
$$T_i = L(32 + 6q)/(13 + 8q), \quad T_d = 4L/(11 + 2q).$$

These yield $K_p \approx 6.730$, $T_i \approx 6.118$ s, and $T_d \approx 0.946$ s. The initial response has pronounced oscillation; subsequent adjustment reduces the visible overshoot. The source does not specify a separate lambda design parameter, so the results are attributed to the stated equations without inferring an unreported design setting.

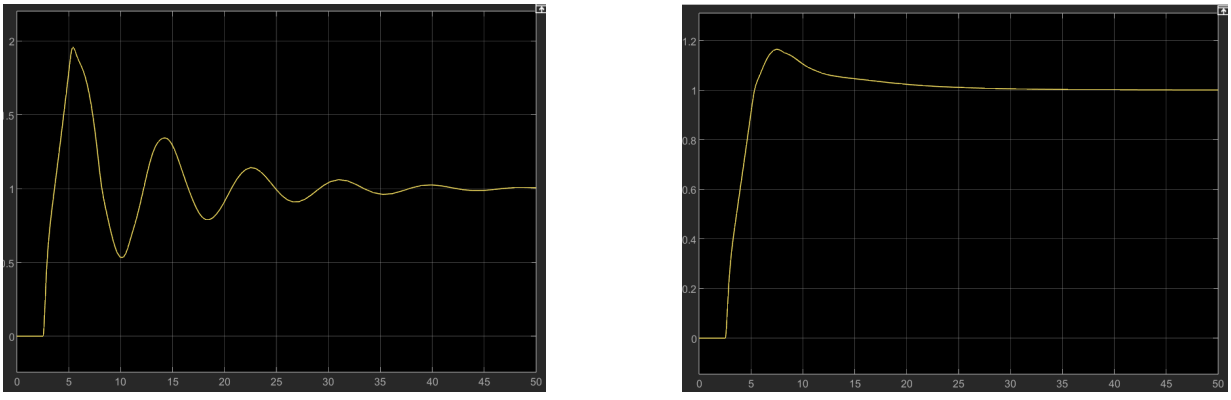


Figure 6. Reported lambda procedure: calculated gains (left, p. 23) and manually adjusted gains (right, p. 24).

Tuning procedure	Calculated (P, I, D)	Adjusted (P, I, D)
Generalized Ziegler-Nichols	(1.3286, 0.1101, 4.0088)	(2.9, 0.25, 2.0)
Reported lambda procedure	(6.730, 1.100, 6.367)	(3.7, 0.4, 3.0)

6. Smith predictor compensation

The final stage investigates model-based compensation of transport delay using a Smith predictor. An internal model provides a delay-free prediction, while the difference between the measured plant output and the delayed model output supplies a correction. With an accurate internal model, this structure reduces the effect of the known delay on the feedback seen by the controller. The physical output still retains the plant transport delay.

Two configurations are considered: one with the complete fourth-order process dynamics and one with the identified three-parameter FOPDT model. Both stored PID blocks use P = 6.730, I = 1.100, D = 6.367, and N = 10. These are the calculated gains from the preceding procedure, rather than its manually adjusted gains.

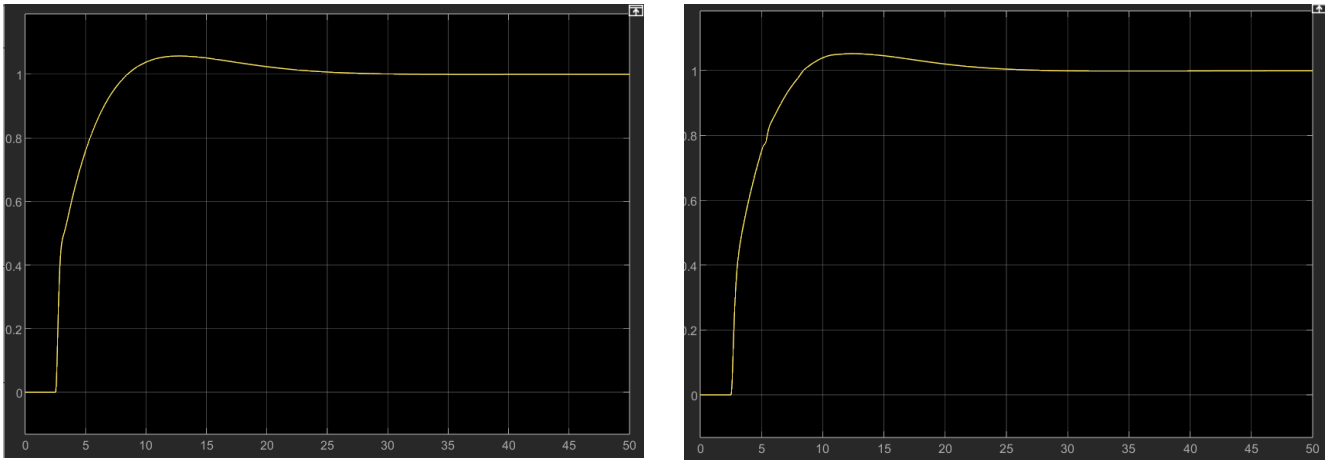


Figure 7. Smith predictor step responses with full process dynamics (left, p. 25) and the three-parameter internal model (right, p. 26).

Both reported responses track the unit reference after the initial delay and show modest visible overshoot followed by convergence toward unity. The similar plotted behavior indicates that the reduced internal model can reproduce the principal tracking characteristics in this nominal simulation. The supplied evidence does not establish equivalent robustness or superior performance under plant-model mismatch.

7. Main findings

Reduced-order modeling. The three-parameter model provides a compact description of the dominant lag and delay. The four-parameter model improves the displayed transient fit and recovers a gain close to one, with a fitted delay near the physical value of 2.5 s.

PID tuning. Relay feedback yields $K_u \approx 6.6845$ and $T_u \approx 9.806$ s. The formula-based designs provide starting points, while the reported manual adjustments improve the visible balance between response speed, overshoot, and oscillation. A single best controller cannot be ranked rigorously without common numerical performance criteria.

Delay compensation. Smith predictor simulations show satisfactory nominal reference tracking with both full and reduced internal models. This supports reduced-order modeling as a practical basis for the investigated compensation structure.

8. Scope of the evidence

This summary reports the supplied simulation results and code settings; it does not present an independent rerun. The report contains no tabulated rise times, settling times, error integrals, control-effort measures, or stability margins. Disturbance rejection, measurement noise, actuator limits, and delay uncertainty are not evaluated. These omissions limit conclusions to nominal step-reference behavior.

Source note. All figures are reproduced from the supplied Persian technical report, with English captions added. Numerical values are transcribed from that report and checked against the two MATLAB scripts and four Simulink models. The original figure labels are retained; “3rd Order” and “4th Order” in Figures 1-2 denote three- and four-parameter models, respectively.